MILITARY SPECIFICATION

DEVICE, RETENTION, FOR CHARGE, PROPELLING, 155MM, XM211

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification covers retention device for use with 155MM Propelling Charge, XM211.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286 - Propellants, Solid, Sampling, Examination and Testing

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8799112 - Marking Diagram and sealing for Inter-Plant or Conus Shipment of Ammunition Fiberboard Boxes

9342673 - Packing & Marking for Box, Fiberboard, for Retention Device for 155MM XM211 Propelling Charge

9340750 - Retention Device

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Materials. - The constituent materials shall comply with the requirements of all applicable drawings and specifications (See 6.3).

3.2 Retention Device - The retention devices shall comply with all requirements specified in Drawing (dwg.) 9340750.

3.3 Stability. - The nitrocellulose in the retention device shall not completely change the color of the methyl violet paper to salmon pink color in less than thirty-five (35) minutes when tested according to 4.4.3.1.

3.4 Weight. - The weight of the retention device shall be 7.6± 0.4 ounces when tested in accordance with 4.4.2.2.

3.5 Cracks and Soft Spots. - All parts shall be free of cracks and soft spots.

3.6 Workmanship. - All parts shall be fabricated in a thorough, workmanship manner. Each retention device shall be free of cracks, pores, discontinuities, sags, blister, pinholes or foreign material.

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3.7 First Article Inspection. - This specification makes provisions for First Article inspection. Requirements for the submission of First Article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the Contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections - The following types of inspection shall be conducted on this tiem:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a First Article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The First Article sample shall consist of 25 retention devices.

4.3.2 Inspection to be Performed. - The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and to the inspection and test of this specification.

4.3.3 Rejection - See MIL-A-48078.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of retention device shall contain:

a.. Nitrocellulose from one manufacturer and from not more than one production lot.

b. Diphenylamine from one manufacturer and from not more than one production lot.

c. Resin from one manufacturer.

d. Kraft from one manufacturer and not more than one production lot.

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4.4.2 Examination. - See MIL-A-48078.

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE ISNPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-63417 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Nitrocellulose				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Stability			3.3	4.4.3.1
Major	None defined				
Minor	None defined				

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-63417 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340750
4.4.2.2	Device, Retention				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cracks and soft spots		100%	3.5	visual
102	Length		100%	3.2	gage
103	Straightness		100%	3.2	gage
104	Weight		0.40%	3.4	4.5.2
105	Composition	10		3.2	4.4.3.2
106	Wall thickness		0.40%	3.2	gage
107	Diameter		0.40%	3.2	gage
Minor					
201	Workmanship		0.65%	3.7	visual

NOTES:

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QUALTIY CONFORMACE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-63417 (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.3	Sealed Barrrier Bag		1 OF 1		9342673
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bag punctured, torn or cut		0.40%	3.2	visual
102	Seal mislocated or incomplete		0.40%	3.2	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMACE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-63417 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9342673
4.4.2.4	Box, Sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Sealing strip torn, incomplete, badly wrinkled or otherwise fails to seal box		0.65%	3.2	visual
202	Box torn, cut or punctured		0.65%	3.2	visual
203	Marking missing, misleading or unidentifiable		0.65%	3.2	visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing

4.4.3.1 Stability (See 3.3). - A stability heat test at 135.5°C shall be performed in accordance with MIL-STD-286, Method 404.1.2. A random sample of 10 retention devices shall be selected from the First Article sample and the first three production lots to provide the required test specimens. Failure of any specimen is cause for lot rejection. When three consecutive lots have been accepted, sampling may be reduced to 5 retention devices from each lot. However, if a failure occurs, sampling shall revert to 10 per lot until 3 consecutive lots have been accepted.

4.4.3.2 Composition. - Ten (10) retention devices shall be randomly selected from the First Article sample and from each of the first three production lots to provide the required test specimens. Failure of any specimen is cause for lot rejection. When three consecutive lots have been accepted, sampling may be reduced to five from each lot. If a failure occurs, sampling shall revert to ten per lot until three consecutive lots have been accepted. Duplicate specimens shall be obtained from each sample for analysis in accordance with 4.5.1.

4.4.4 Inspection Equipment. - The inspection equipment required to perform the examination and tests prescribed herein is described in the Paragraph Reference/Inspection Method Column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078.

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4.5 Test Methods and Procedures

4.5.1 Composition

4.5.1.1 Preparation of Specimen (See 4.3.3.3). - Each specimen shall be divided into 1/8 inch (approximate) cubes or equivalent. The specimens shall then be thoroughly mixed to form a composite. From the composite, approximately 1.0 g shall be selected as the analysis sample and weighed to the nearest 0.2 milligram (mg), (Weight A) in an extraction thimble. The same thimble should be carried through the extractions in 4.5.1.3.1 and 4.5.1.3.3.

4.5.1.2 Total Volatiles (Informational). - The samples from 4.5.1.1 shall be dried for 3 hours at 100 ± 2°C, cooled in a desiccator (1/2 hour min.), and weighed to the nearest 0.2 mg (Weight W). Calculate the percent total volatiles as follows:

$$\% \text{ Total Volatiles} = \frac{A - W \times 100}{A}$$

4.5.1.3 Analytical Procedure

4.5.1.3.1 Extraction. - Using a tared, extra-coarse sintered glass thimble, extract each sample from 4.5.1.2, with methylene cloride, for a minimum of 16 hours in a Roweg Extractor (MIL-STD-286, Method 104.1). Evaporate the extract to dryness, using a current of clean, dry air. This extract is to be used for diphenylamine determination in 4.5.1.3.2. Dry the residues (nonextractable material) in the extraction thimbles by aspirating the thimble and contents until the odor of methylene chloride is not discernible. Dry the thimble plus residue for 1/2 hour minimum at 100 ± 2°C. Cool in a desiccator (1/2 hour min.). Determine the dry weight of this residue (Weight C) to the nearest 0.2 mg as the difference between this weight and the weight of the empty thimble. This residue is to be used for the nitrocellulose determination in 4.5.1.3.3.

4.5.1.3.2 Diphenylamine (DPA). - To the dried extract from 4.5.1.3.1, add 50 milliliters (ml) of glacial acetic acid. If necessary, warm slightly to dissolve the extract, then cool to room temperature. Complete the determination of diphenylamine in the solution according to Method 201.1 (starting with paragraph 5.5) of Standard MIL-STD-286, except that 25 ml of

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methylene chloride is to be added after addition of the potassium iodide solution. When the end point appears to have been reached, (absence of blue color), the flask is stoppered and shaken. If no color appears, the end point has been reached; otherwise, continue titration until the blue color is discharged and shaking produces no further color. Calculate the percent DPA as shown in Standard MIL-STD-286, Method 201.4, paragraph 5.11.

4.5.1.3.3 Nitrocelllulose. - To provide good heat conduction to the solvent, place a sufficient quantity of glycerin to the flask bottom and the surface of the heat source to produce a continuous film. Add 10 ml of ethyl alcohol to the thimble and residue from 4.5.1.3.1 and extract with acetone using the method in 4.5.1.3.1. Evaporate the acetone extract to dryness and add 25 ml of n-butyl acetate and 45 ml of glacial acetic acid. Replace the air with a stream of inert gas (CO_2 or N_2). Maintain the inert atmosphere throughout this test. Reflux the solution for five (5) minutes. Add to the flask, via the condenser, 25 ml of 0.7N ferrous ammonium sulfate (prepared per MIL-STD-286, Method 604.1) and 10 ml concentrated hydrocloric acid. Reflux for thirty (30) minutes more to obtain a yellow liquid in which the specimen is completely disintegrated. With the inert gas still running, cool in a water bath to room temperature. Add 5 ml of 20% ammonium thiocyanate solution (MIL-STD-286, Method 601.1). Titrate with a 0.2N titanous chloride solution (MIL-STD-286, Method 601.1). Perform a blank titration on the reagents. Calculate the percent NC as follows:

$$\% \text{ NC} = \frac{46.69N\ (V-B)\ X\ 100}{E\ W}$$

Where:

V = Volume titanous chloride use for sample, ml
B = Volume titanous chloride used for blank, ml
W = Weight of dried sample, gm
N = Normality of titanous chloride
E = Percent nitrogen in the nitrocellulose

4.5.1.3.4 Kraft and Resin . - Calculate the percent kraft and resin as follows:

% Kraft + % Resin = 100 - (%NC + %DPA)

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4.5.2 Weight of Retention Device. - The sample retention devices shall be dried in an oven (see 6.5) maintained at 160 degrees Fahrenheit (^{o}F) $\pm$ 10°F for a period of 2 hours. The retention devices shall be removed from the oven and immediately place each retention device in a tared container (the container may be a plastic (polyethylene) bag with a metal or plastic tie) and allowed to cool for 1 hour in the same room as the balance. Weigh to the nearest 0.01 gram. If the sample fails due to high weights, repeat the above procedure except that the oven time shall be 0.5 hours. Samples may be dried as many times as necessary or until a constant weight is obtained. The weight shall be determined as follows:

Weight of Retention Device = P - Q
Q - Weight of container
P - Weight of container plus retention device

5. PACKAGING

5.1 Packing, Level C. - All parts shall be packed in accordance with dwg 9342673.

5.2 Marking. - Marking shall be in accordance with dwg. 8799112.

6. NOTES

6.1 Ordering Data. - See MIL-A-48078.

6.2 Submission of Designs for Approval. - See 6.2.3. of MIL-A-48078. Submit equipment deisgns, as required to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.3 Submission of Ammunition Data Cards. - In addition to the normal distribution of records, when the retention devices are procured by the Army, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
US Army Armament Research and Devleopment Command
ATTN: DRDAR-QAR-R
DRDAR-LCA-G
Dover, NJ 07801

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6.4 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command ARRADCOM may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizence of ARRADCOM.

6.5 Ovens. - Ovens utilized for drying retention devices to determine weight will comply with PAD 1006.

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